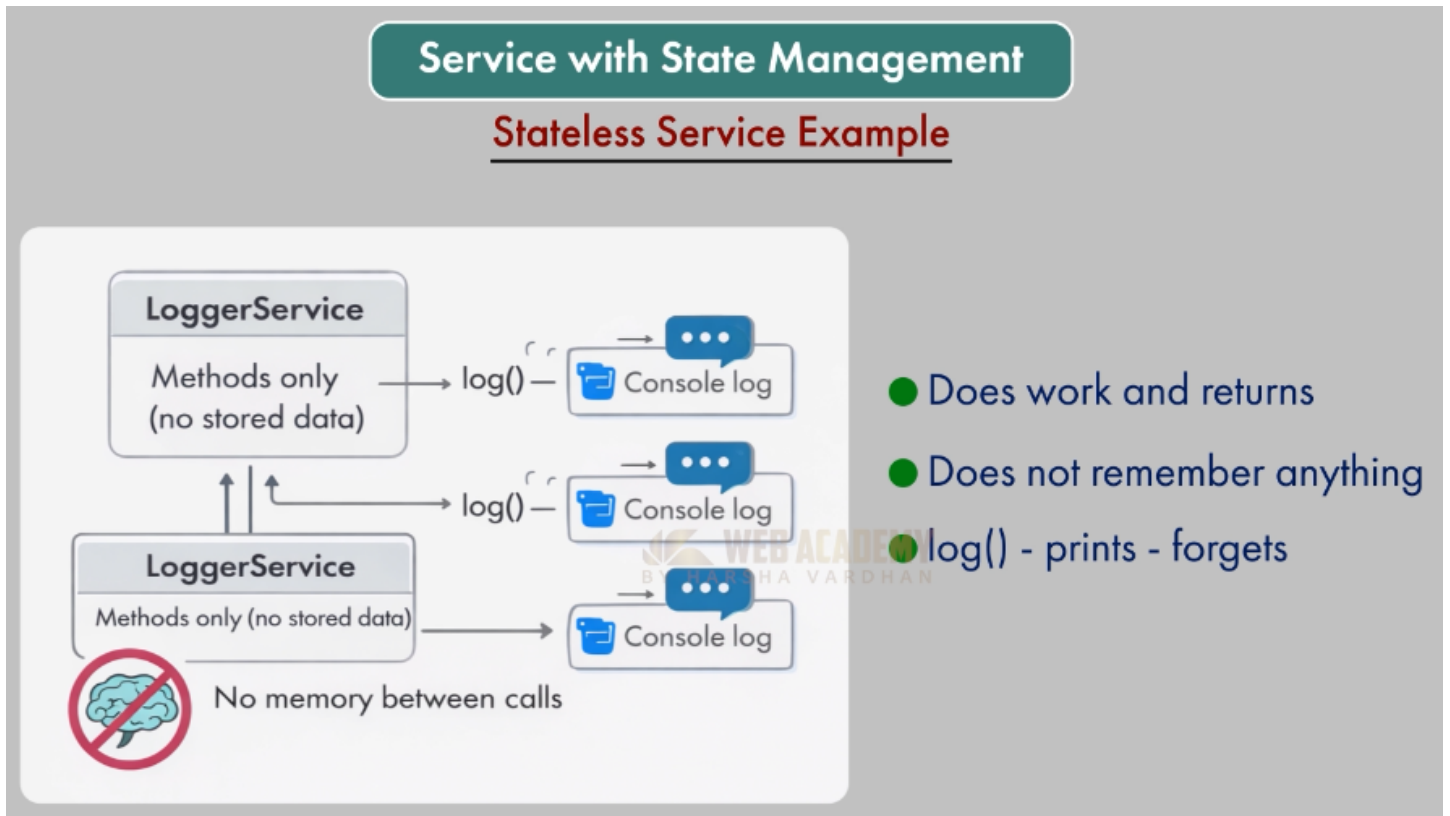


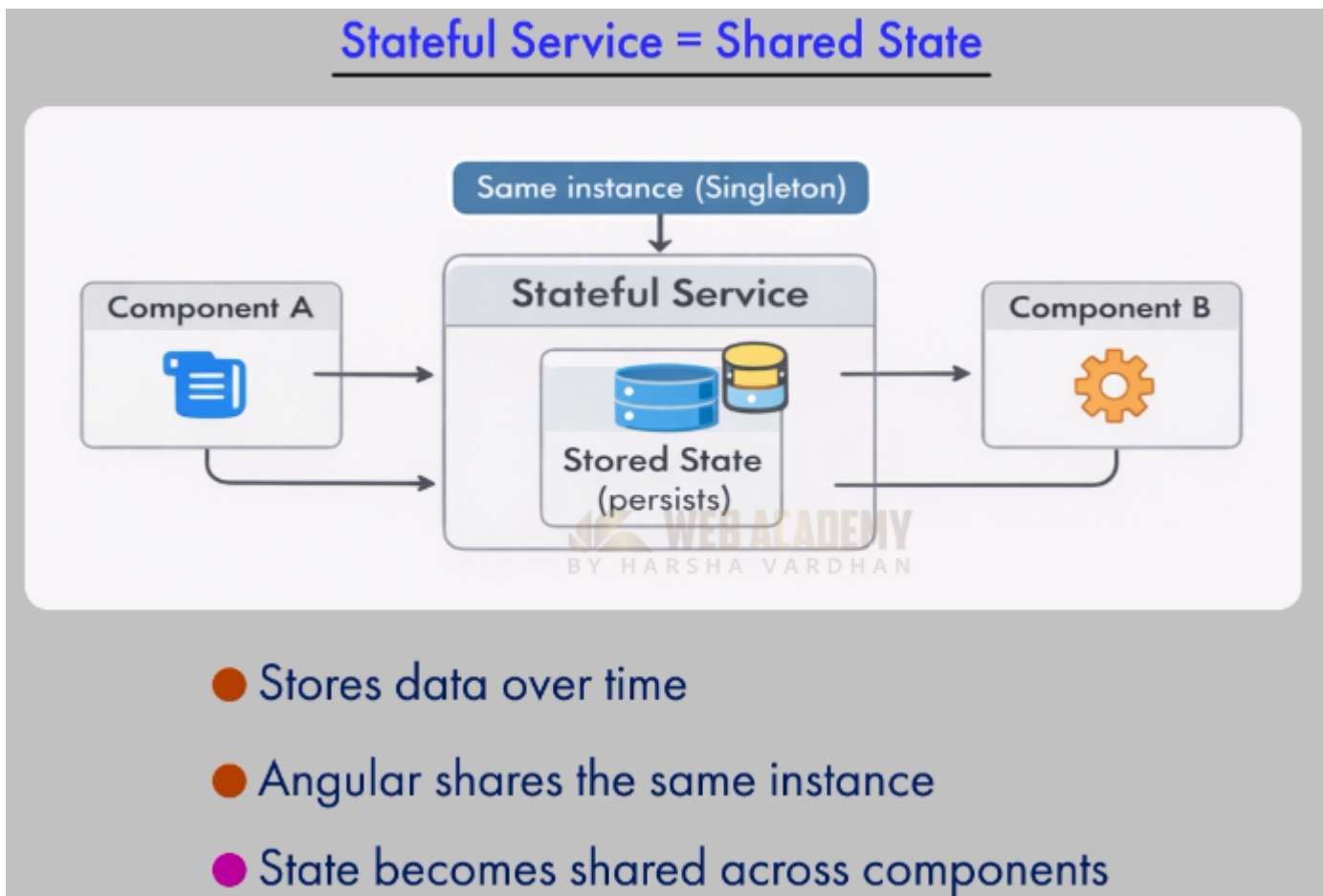
Stateless Service

No memory between calls



វាមិនរក្សាទិន្នន័យនោះទេ (cannot share data)

Stateful Service = Shared State



Create Counter to remember the count:

>> ng g s services/counter

```
import { Injectable } from '@angular/core';

@Injectable({
  providedIn: 'root',
})
export class Counter {
}
```

⇒ Then

```
import { Injectable } from '@angular/core';

@Injectable({
  providedIn: 'root',
})
export class Counter {

  private count: number = 0;
}
```

យើងប្រើប្រាស់ private keyword ព្រោះដើម្បីកុំអោយ component ធ្វើការកែប្រែបាន

- Add the method getCount() and increstment method to count by 1

```
TS counter.ts X
src > app > services > TS counter.ts > Counter > reset
1  import { Injectable } from '@angular/core';
2
3  @Injectable({
4    providedIn: 'root',
5  })
6  export class Counter {
7
8    private count: number = 0;
9
10   getCount():number {
11     return this.count;
12   }
13
14   increment(): void {
15     this.count++;
16   }
17
18   decrement(): void {
19     this.count--;
20   }
21
22   reset(): void {
23     this.count = 0;
24   }
25
26 }
27
```

2- Create components to use the service count

>> ng g c counter-display

>> ng g c counter-controls

Service Demo

Counter Display

0

Counter Controls



- Counter service

```
TS counter-display.ts X TS counter.ts X
src > app > services > TS counter.ts > ...
 1  import { Injectable } from '@angular/core';
 2
 3  @Injectable({
 4    providedIn: 'root',
 5  })
 6  export class CounterService {
 7
 8    private count: number = 0;
 9
10    getCount(): number {
11      return this.count;
12    }
13
14    increment(): void {
15      this.count += 10;
16    }
17
18    decrement(): void {
19      this.count -= 5;
20    }
21
22    reset(): void {
23      this.count = 0;
24    }
25
26  }
27
```

- TS in the counter control

```
src > app > counter-controls > <> counter-controls.html > div.counter-controls
Go to component
1  <div class="counter-controls">
2    <h3>Counter Controls</h3>
3    <div class="buttons">
4      <button class="btn-decrement" (click)="onDecrement()">-</button>
5      <button class="btn-increment" (click)="onIncrement()">+</button>
6      <button class="btn-reset" (click)="onReset()">Reset</button>
7    </div>
8  </div>
```

```
TS counter-display.ts X TS counter-controls.ts X
src > app > counter-controls > TS counter-controls.ts > CounterControls
1  import { Component } from '@angular/core';
2  import { CounterService } from '../services/counter';
3
4  @Component({
5    selector: 'app-counter-controls',
6    imports: [],
7    templateUrl: './counter-controls.html',
8    styleUrls: ['./counter-controls.css'],
9  })
10 export class CounterControls {
11
12   constructor(private counterService: CounterService){
13
14   }
15
16   onIncrement() : void{
17     this.counterService.increment();
18   }
19
20   onDecrement() : void{
21     this.counterService.decrement();
22   }
23
24   onReset(): void {
25     this.counterService.reset();
26   }
27
28 }
29
```

- Ts control display

```
rc > app > counter-display > TS counter-display.ts > ...
1  import { Component } from '@angular/core';
2  import { CounterService } from '../services/counter';
3
4  @Component({
5    selector: 'app-counter-display',
6    imports: [],
7    templateUrl: './counter-display.html',
8    styleUrls: ['./counter-display.css'],
9  })
10 export class CounterDisplay {
11
12   constructor(private counterService: CounterService){}
13
14   getCurrentCount(): number{
15     return this.counterService.getCount();
16   }
17
18 }
19
```

Control service counter

Counter Demo

Counter Display
0

Counter Status
Zero

Counter Controls

-

+

Reset

- We create one control service then for the component can inject for using this service.

```

export class CounterStatus {

  constructor(private counterService:CounterService){}

  getStatus():string
  {
    const count = this.counterService.getCount();

    if(count<0) return 'Negative';
    if(count === 0) return 'Zero';
    if(count <=5) return 'Low';
    if(count <=10) return 'Medium';
    return 'Hight';
  }

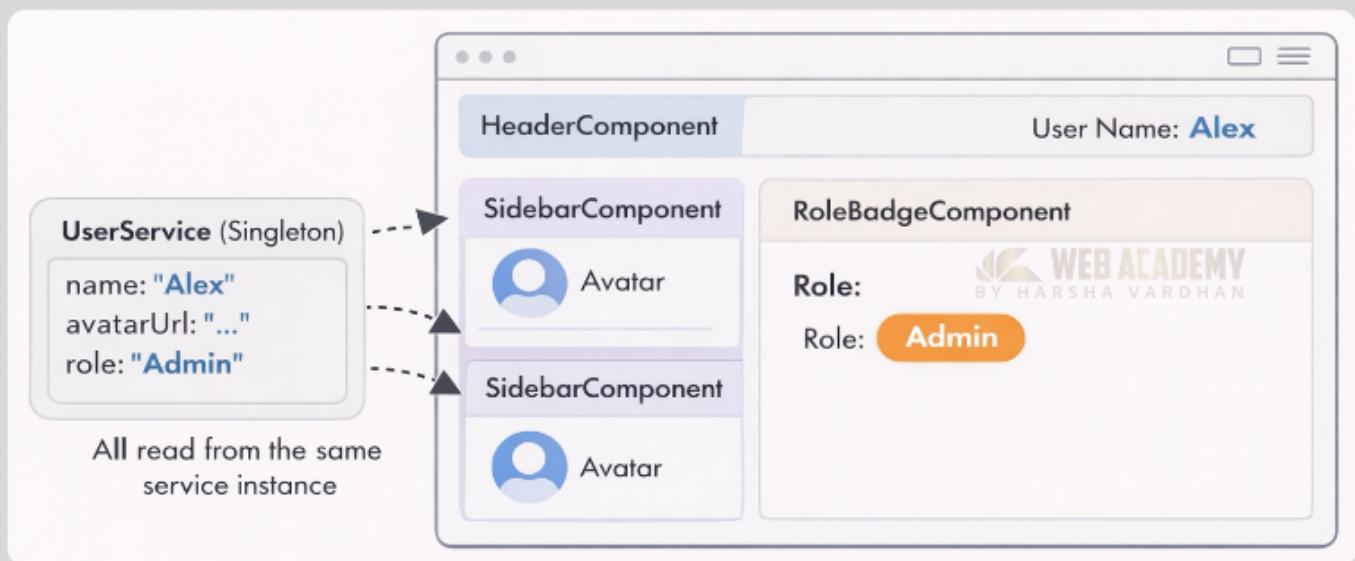
  getStatusColor():string {

    const count = this.counterService.getCount();

    if (count < 0) return '■#e74c3c';
    if (count == 0) return '■#95a5a6';
    if (count <= 5) return '■#3498db';
    if (count <= 10) return '■#f39c12';
    return '■#27ae60';
  }
}

```

Real apps: multiple UI areas read the same UserService

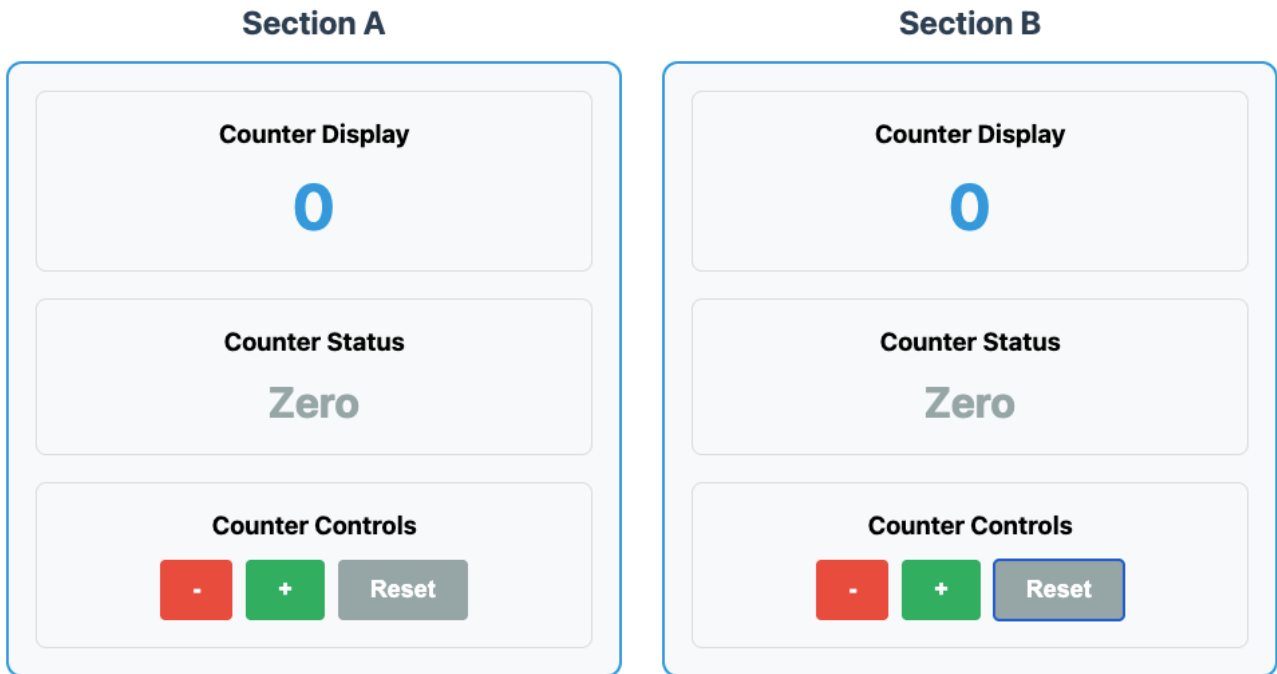


When **UserService** data changes - all UI updates automatically

Separate the Counter

យើងចង់បានអោយ counter independence and count depend on each component

Services Demo



- We create on wraper class to for the counter:

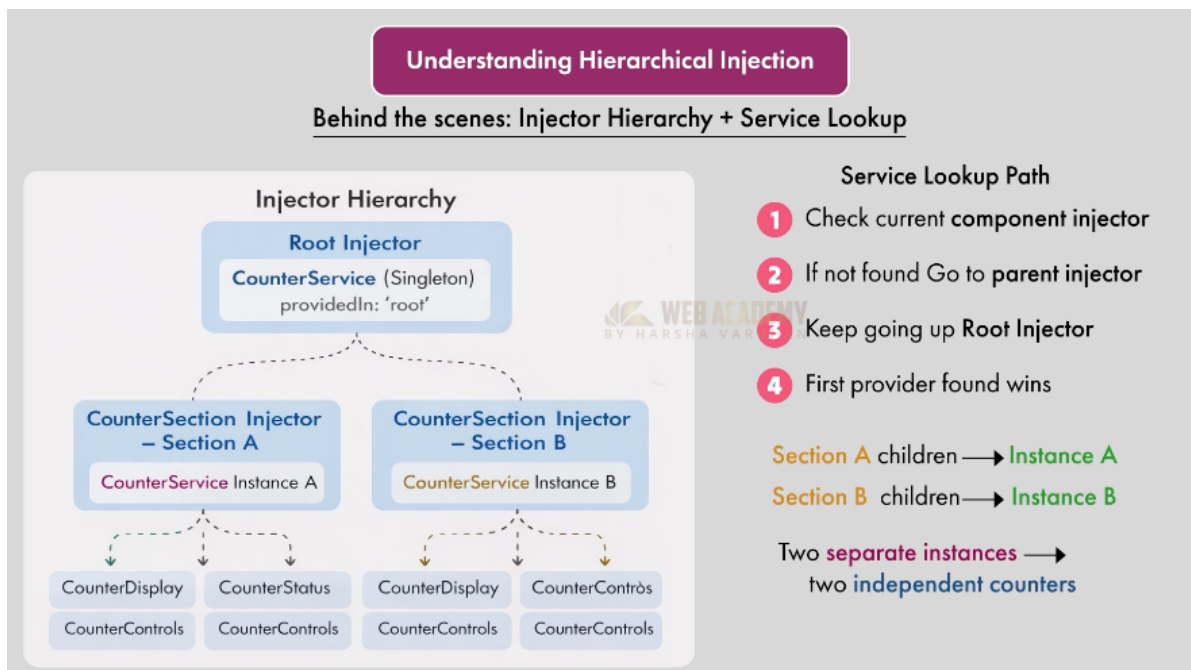
>> ng g c counter-section

```
<> counter-status.html  TS counter-section.ts  <> counter-section.html  # cou
src > app > counter-section > TS counter-section.ts > CounterSection
1  import { Component } from '@angular/core';
2  import { CounterDisplay } from '../counter-display/counter-display';
3  import { CounterControls } from '../counter-controls/counter-controls';
4  import { CounterStatus } from '../counter-status/counter-status';
5  import { CounterService } from '../services/counter';
6
7  @Component({
8    selector: 'app-counter-section',
9    imports: [CounterDisplay,CounterControls,CounterStatus],
10   templateUrl: './counter-section.html',
11   styleUrls: ['./counter-section.css'],
12   providers:[CounterService]
13 })
14 export class CounterSection {
15
16 }
17
```

In the html we use :

```
providers:[CounterService]
```

ដើម្បីធ្វើការ កំណត់ scope របស់ component counter-section រាល់ដងសំរាប់ service the root is the scrop when it as for the state it will as for the root scope so when we put the provider in the component it will us the component for his memory.



```
<div class="counter-section">
  <app-counter-display></app-counter-display>
  <app-counter-status></app-counter-status>
  <app-counter-controls></app-counter-controls>
</div>
```

Services Demo

Section A

Counter Display

2

Counter Status

Low

Counter Controls

-
+
Reset

Section B

Counter Display

5

Counter Status

Low

Counter Controls

-
+
Reset